

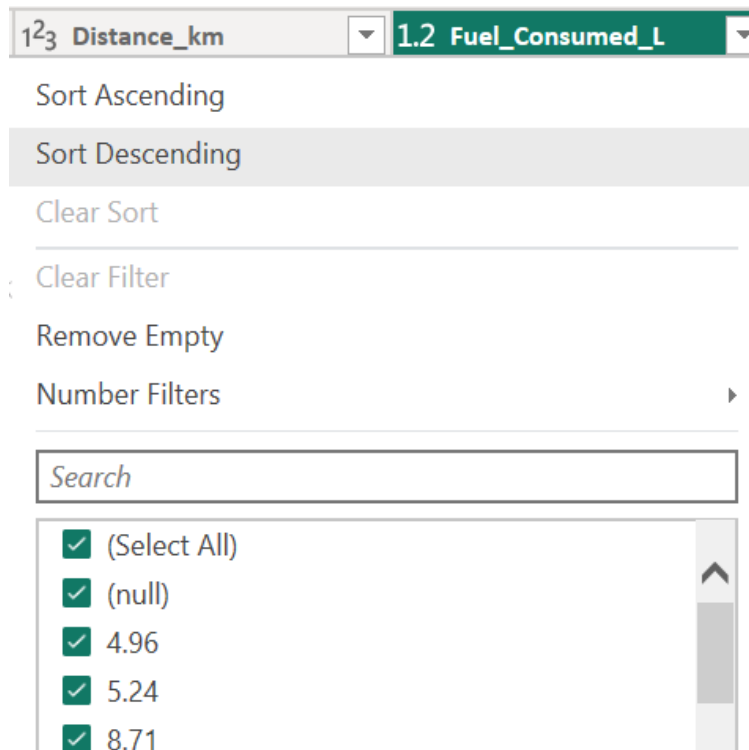
Project: Power BI Module End Project

Scenario: A logistics company wants to analyze its fleet performance in terms of on-time deliveries, fuel efficiency, and cost per mile.

- First, I copied the Trip Data and Vehicle Data into my own worksheet.
- I created tables and assigned the names 'Trip Table' to the trip data and 'Vehicle Table' to the vehicle data for clarity.
- After importing the data into Power BI, I used Power Query to clean and prepare the data for analysis.

1. Data Cleaning & Modeling:

- Fix missing fuel consumption values (use mean imputation).
- Here, the missing values in the Fuel Consumption column are represented as null.



- I computed the mean value using Transform → Statistics → Average, and then filled the missing values in the column by replacing the nulls with this average.

Queries [2]

1²₃ Trip_Data

Vehicle_Master

✕

✓

fx

= List.Average(#"Changed Type"[Fuel_Consumed_L])

91.882978723404264

= Table.TransformColumnTypes(#"Promoted Headers",{{"Trip_ID", type text}, {"Vehicle_ID", type text},

AB_C Destination

1²₃ Distance_km

1.2 Fuel_Consumed_L

AB_C Delivery_Status

Replace Values

Replace one value with another in the selected columns.

Value To Find

null

Replace With

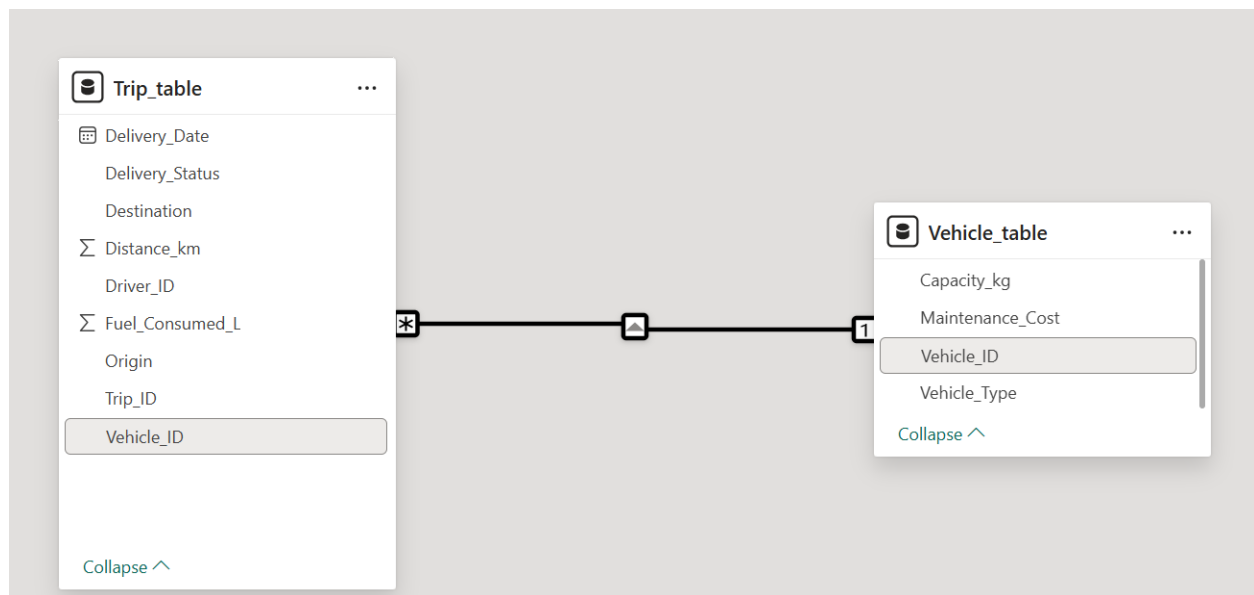
91.88

OK

Cancel

Bangalore	1524	104.51	On-Time
Mumbai	1956	179.88	On-Time
Mumbai	858	null	Late
Kolkata	1269	102.91	On-Time
Delhi	1565	107.23	On-Time
Pune	1796	155.52	On-Time
Bangalore	1540	148.87	On-Time

- Create a relationship using Vehicle_ID between Trip_Data and the Vehicle Master table.



2. DAX Measures:

- Fuel Efficiency = Distance / Fuel Consumed

DAX Formula:

Fuel Efficiency = `DIVIDE( SUM(Trip_table[Distance_km]), SUM(Trip_table[Fuel_Consumed_L]))`

The screenshot shows the Power BI interface. On the left, the DAX formula bar contains the following code:

```
1 Fuel Efficiency =
2 DIVIDE(
3     SUM(Trip_table[Distance_km]),
4     SUM(Trip_table[Fuel_Consumed_L])
5 )
```

Below the formula bar, a message reads: "Build visuals with your data. Select or drag fields from the Data pane onto the report canvas." Below this message are icons for a table, a bar chart, and a line chart.

On the right, the "Visualizations" pane shows a "Build visual" section with various chart icons. The "Data" pane shows the "Trip_table" with the following fields listed: Delivery_Date, Delivery_Status, Destination, Distance_km, Driver_ID, Fuel_Consumed_L, and Fuel Efficiency. The "Fuel Efficiency" field is highlighted in the list.

- On-Time Delivery % = On-Time Trips / Total Trips:

DAX Formula:

On-Time Trips = CALCULATE(COUNTROWS(Trip_table),
Trip_table[Delivery_Status] = "On-Time")

The screenshot shows the Power BI Desktop interface. The formula bar contains the following DAX formula:

```
1 On-Time Trips =
2 CALCULATE(
3     COUNTROWS(Trip_table),
4     Trip_table[Delivery_Status] = "On-Time"
5 )
6
```

The Data pane on the right shows the 'Trip_table' with the following fields: Delivery_Date, Delivery_Status, Destination, Distance_km, Driver_ID, Fuel Efficiency, Fuel_Consumed..., and On-Time Trips. The 'On-Time Trips' field is highlighted.

The Visualizations pane on the left shows a 'Build visual' button and a list of visualization types. The 'Build visual' button is highlighted.

The report canvas shows a message: 'Build visuals with your data. Select or drag fields from the Data pane onto the report canvas.'

Total Trips = COUNTROWS(Trip_table)

The screenshot shows the Power BI Desktop interface. The formula bar contains the following DAX formula:

```
1 Total Trips = COUNTROWS(Trip_table)
2
```

The Data pane on the right shows the 'Trip_table' with the following fields: Delivery_Date, Delivery_Status, Destination, Distance_km, Driver_ID, Fuel Efficiency, Fuel_Consumed..., On-Time Delivery..., On-Time Trips, Origin, and Total Trips. The 'Total Trips' field is highlighted.

The Visualizations pane on the left shows a 'Build visual' button and a list of visualization types. The 'Build visual' button is highlighted.

The report canvas shows a message: 'Build visuals with your data. Select or drag fields from the Data pane onto the report canvas.'

On-Time Delivery % = DIVIDE([On-Time Trips], [Total Trips], 0)

The screenshot shows the Power BI Desktop interface. The formula bar contains the following DAX formula:

```
1 On-Time Delivery % =
2 DIVIDE([On-Time Trips], [Total Trips], 0)
3
```

The Data pane on the right shows the 'Trip_table' with the following fields: Delivery_Date, Delivery_Status, Destination, Distance_km, Driver_ID, Fuel Efficiency, Fuel_Consumed_L, and On-Time Delivery %. The 'On-Time Delivery %' field is highlighted.

The Visualizations pane on the left shows a 'Build visual' button and a list of visualization types. The 'Build visual' button is highlighted.

The report canvas shows a message: 'Build visuals with your data. Select or drag fields from the Data pane onto the report canvas.'

The name bar at the bottom shows the name of the measure: 'On-Time Delivery %'.

- Cost per km = (fuel cost + Maintenance Cost) / Distance

DAX Formula:

Cost per km = DIVIDE(100 + SUM(Vehicle_table[Maintenance_Cost]),
SUM(Trip_table[Distance_km]))

The screenshot shows the Power BI Desktop interface. The formula bar at the top contains the DAX formula for 'Cost per km':

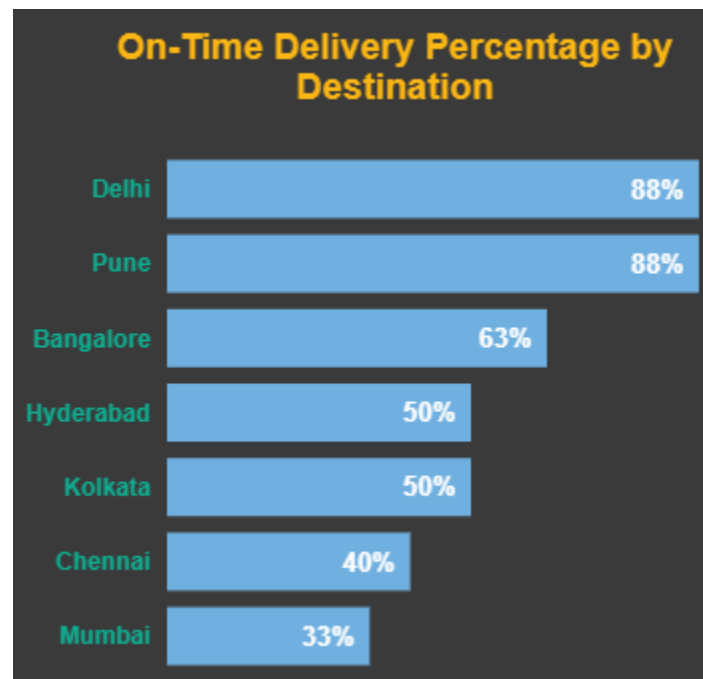
```
1 Cost per km =
2 DIVIDE(
3     100 + SUM(Vehicle_table[Maintenance_Cost]),
4     SUM(Trip_table[Distance_km])
5 )
```

Below the formula bar, the 'Build visuals with your data' section is visible, along with the 'Data' pane on the right. The 'Data' pane shows the following fields:

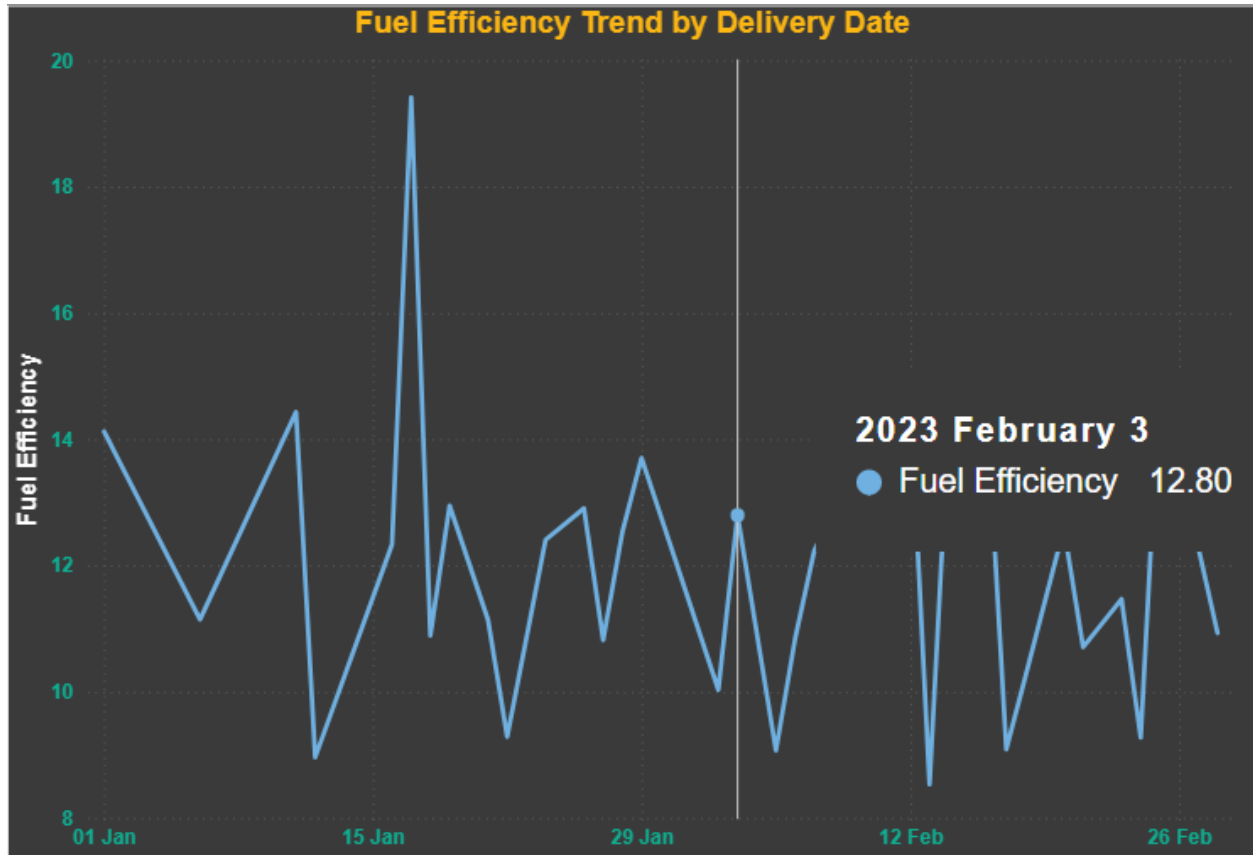
- Trip_table
- Vehicle_table
 - Capacity_kg
 - Cost per km (selected)
 - Maintenance_Cost
 - Vehicle_ID
 - Vehicle_Type

3. Visualization:

- Bar chart: On-Time Delivery % by Destination.
 - Y-axis → Destination
 - X-axis → On-Time Delivery%



- Line chart: Fuel efficiency trend by delivery date.
 - Y-axis→Delivery_Date
 - X-axis→Fuel Efficiency



Cards visualization:

- Avg. Delivery Time

DAX Formula:

Delivery Time (hrs) = $\text{DIVIDE}(\text{Trip_table}[\text{Distance_km}], 40)$

Avg Delivery Time = $\text{AVERAGE}(\text{Trip_table}[\text{Delivery Time (hrs)}])$

```

1 Delivery Time (hrs) =
2 DIVIDE(Trip_table[Distance_km], 40)
3

```

Visualizations

Build visual

Quick meas

Filters

Name 'Trip_table[Delivery Time (hrs)]'

Data

Search

✓ Trip_table

- ☐ Avg Delivery Time
- ☐ Delivery Time (hrs)
- ☐ Delivery_Date
- ☐ Delivery_Status
- ☒ Destination
- ☐ Σ Distance_km



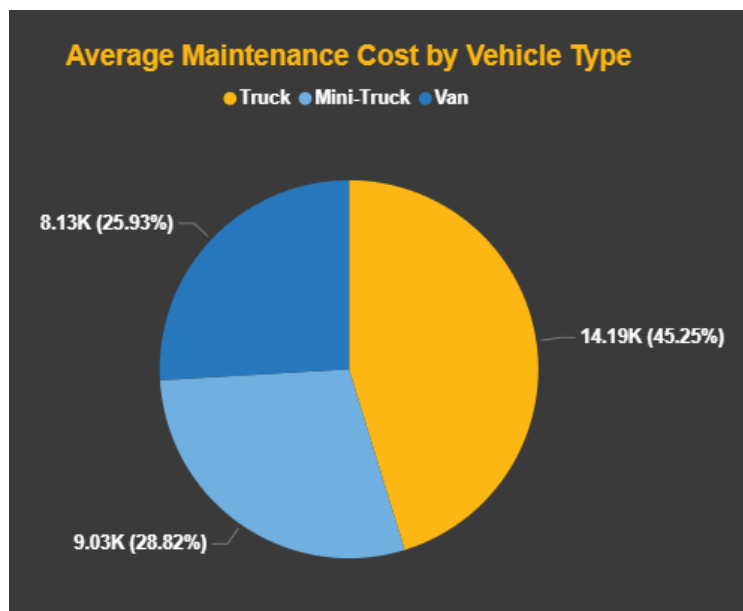
- Average cost per km

DAX Formula:

Average Cost per km = AVERAGEX(Trip_table, [Cost per km])



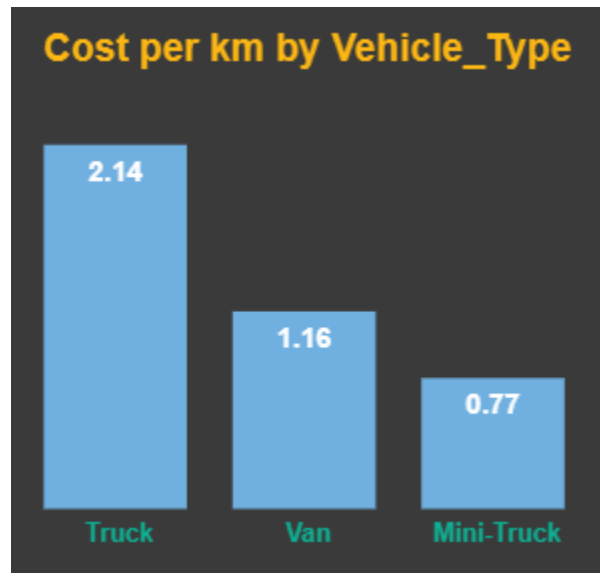
- Pie chart: vehicle type vs Average maintenance cost
 - Legend→Vehicle_Type
 - Values→Average Of Maintenance Cost



4. AI-Powered Visuals

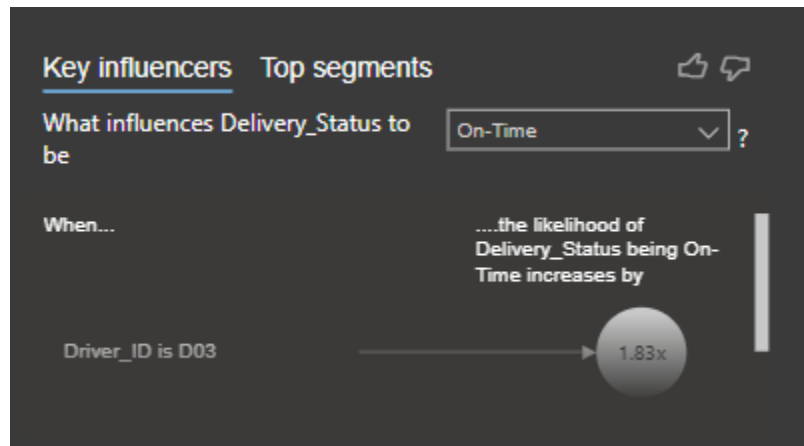
❖ Q&A Visual: ➤ Add Q&A visual → Prompt: “Average Cost per km by vehicle type?”

- Since the Q&A option was not available for me, I used a clustered column chart instead.
- Y-axis→Vehicle_Type
- X-axis→Cost per km



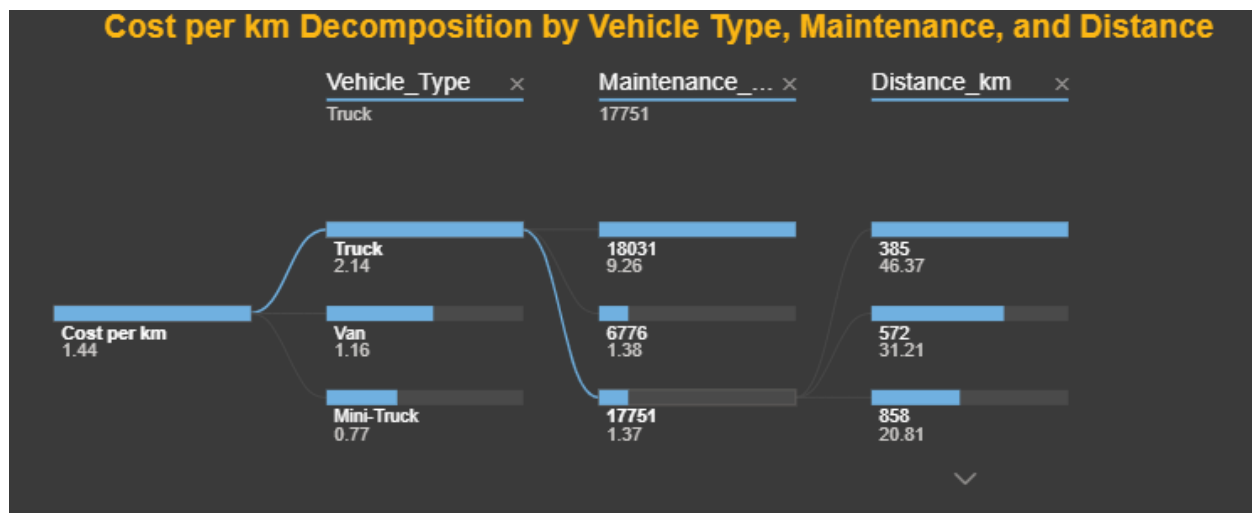
❖ Key Influencers Visual: ➤ Add Delivery Status in the Analyze and explain by -distance in km,vehicle_type, Driver_ID

- Insert Tab→AI Visuals→Key influencers
- Analyze→Delivery_Status
- Explain by→Distance_km, Vehicle_Type and Driver ID



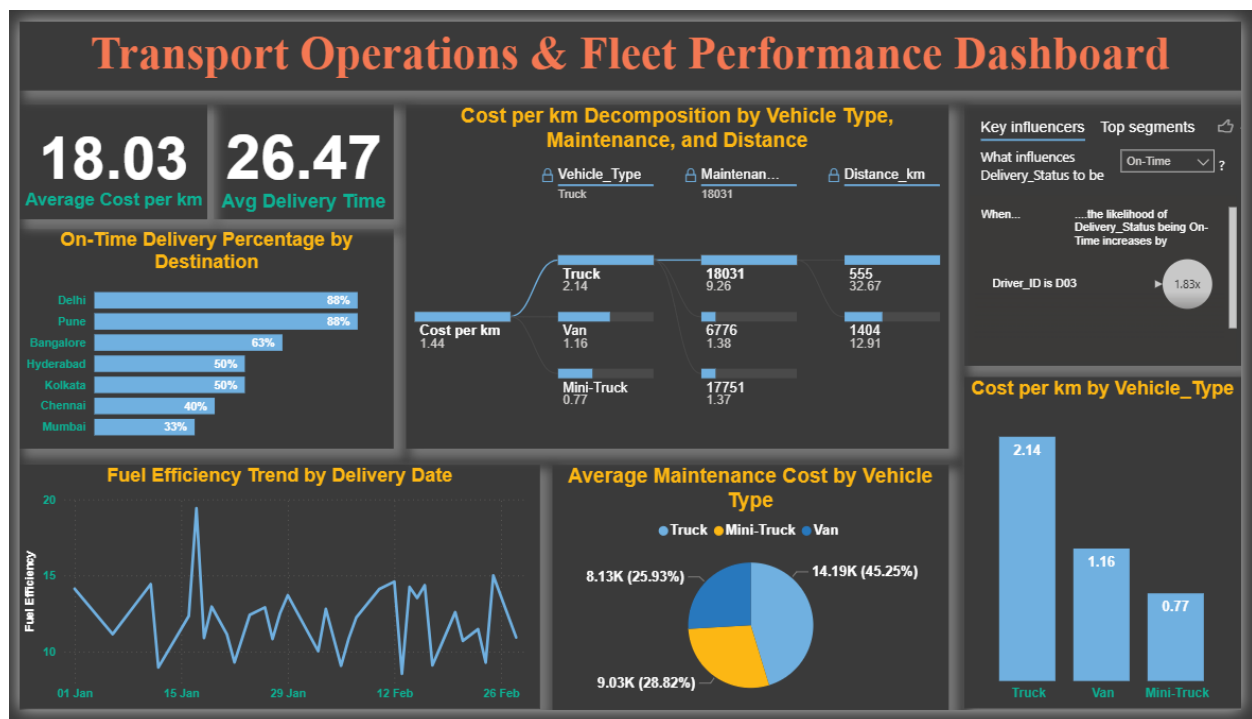
❖ Decomposition Tree (AI Visual): ➤ Analyze Cost per km → explained by Vehicle Type, Maintenance_cost, and Distance_km.

- Insert Tab→AI Visuals→Decomposition tree
- Analyze→Cost per km
- Explain by→Distance_km, Vehicle_Type and Maintenance_Cost



Expected Output: A transport operations dashboard to optimize routes and fleet usage.

Finally, after creating all the charts, I made small changes to the colors and fonts and fixed their placement. The dashboard was successfully created.



THANK YOU